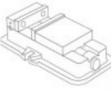
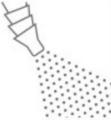
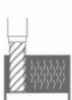


铣刀加工时的要点
Factors for End Mill Operation

铣刀加工时所遇到问题与解决方案
Hardened Steels For Machining Endmills

要点 Factor	注意事项 Instruction and Advice
机床刚性 Rigidity of Machine 	①尽可能用高刚性的设备进行加工。 Use a rigid machine. ②刚性不足时，调整切深量及切削参数。 Adjust cutting conditions according to the rigidity of machine.
夹具和铣刀的偏摆 Collet Chuck and Runout of End Mill 	①请使用高刚性高精度铣刀刀柄。 Use a rigid and precise collet chuck. ②上刀时，尽量将刀具的偏摆控制到最小。 Minimize the run out of end mill.
工件夹钳 work Clamp 	①工件夹具必须固定牢固。 Work piece must be firmly clamped. ②无法改善夹持状态时，调整切深量等切削参数。 In case work piece cannot be firmly clamped, relieve cutting condition.
切削油和切屑 的处理 Cutting Fluid and Chips 	①使用的切削油量须够充足。 Give a sufficient cutting fluid. ②重切削时，推荐使用水溶性切削油。 Recommend water soluble fluid for heavy cutting. ③请注意部分产品为干式切削专用。 Some end mills apply dry cutting only. ④干式切削请使用吹气冷却方式。 Use air blow for dry cutting. ⑤应避免切屑妨碍切削油供应或切削加工。 Remove chips from working area.
铣刀种类的选择 Selection of End Mill 	①请根据不同的加工工件和加工形状，选择适合的铣刀。 Most suitable end mills according to work material and dimension. ②请参照卷头的索引。 Refer to the index table on front page.
切削参数 Cutting Conditions vc vf ap ae 	①请参阅切削参数参考表。 Refer to recommended milling condition table. ②请根据设备的刚性和工件的夹持状态，调整切削参数。 It is necessary to adjust conditions according to the machine rigidity and clamping condition of work material.
铣刀的伸出量 Overhang of End Millfrom Tool Holder 	①在不造成干涉的情况下，请尽量减少刀具的伸出量。 Overhang of end mill must be as short as possible from tool holder. ②当刀具伸出量过长时，请调整切深量及进给速度等切削参数。 In case overhang cannot be shorten, relieve cutting condition .

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问题现象 Symptoms of troubles	原因 Cause	建议措施 Solution
切削时的振刀 Chattering 	主轴转速太快。 进给速度太快。 有效长超出需要，伸出量太长。 加工工件夹持力不足。 切削刀刃磨损过大。 刀具偏摆精度不足。 Excessive spindle speed. Excessive feed. Excessive long of effective length or overhang of end mill. Work material is not firmly clamped. Wear of cutting edge progressed. Excessive chucking runout.	降低主轴转速。 降低进给速度。 有效长和伸出量请调整到最低限度。 牢固固定加工工件。 使用新的刀具或者进行返修。 调整刀具偏摆精度。 Reduce spindle speed. Reduce feed. Adjust effective length and overhang as short as possible. Clamp work material firmly. Use new end mill or regrind. Adjust chucking runout.
切削时的折损 Breakage of End Mill 	切深量太大。 切屑堵塞。 每刃进给量太大。 切削刀刃磨损过大。 Excessive depth of cut. Chips clogged Excessive feed per tooth. Wear of cutting edge progressed.	降低切深量。 调整冷却液的对准方向，使切屑排出顺利。 降低每刃进给量。 使用新的刀具或者进行返修。 Reduce depth of cut. Adjust coolant nozzle to right direction to dispose chips. Reduce feed per tooth. Use new end mill or regrind.
切削中的崩刃 (微小欠角) Chipping of Cutting Edge 	切深量太大。 进给速度太快。 加工工件夹持力不足。 主轴转速太快。 有效长超出需要，伸出量太长。 切削刀刃磨损过大。 产生积屑瘤。 切削时的急冷所造成的损伤。 Excessive depth of cut. Excessive feed. Work material is not firmly clamped. Excessive spindle speed. Excessive long of effective length or overhang of end mill. Wear of cutting edge progressed. Built up edge. Excessive cooling.	降低切深量。 降低进给速度。 牢固固定加工工件。 降低主轴转速。 有效长和伸出量请调整到最低限度。 使用新的刀具或者进行返修。 使用最合适的涂层刀具。 使用吹气冷却或者油雾冷却方式。 Reduce depth of cut. Reduce feed. Clamp work material firmly. Reduce spindle speed. Adjust effective length and overhang as short as possible. Use new end mill or regrind. Choose appropriate coating. Use air blow or oil mist.
切削刃的异常磨损 Abnormal ear 	主轴转速太快。 进给速度太慢。 Excessive spindle speed Too low feed	降低主轴转速。 调高进给速度。 Reduce spindle speed Increase feed
切屑堵塞、熔敷 Clogging and Depositing 	排屑不畅。 进给速度太快。 切深量太大。 刀具刃数过多。 切削刀刃磨损过大。 Chips are not well disposed Excessive feed Excessive depth of cut Inappropriate number of flute Wear of cutting edge progressed	调整冷却液的对准方向，使切屑排出顺利。 降低进给速度。 降低切深量。 选用刃数较少的刀具。 使用新的刀具或者进行返修。 Adjust coolant nozzle to right direction to dispose chips . Reduce feed Reduce depth of cut Use fewer flutes end mill Use new end mill or regrind
切削时的倾倒 Deflection of End Mill 	进给速度太快。 切深量太大。 有效长超出需要，伸出量太长。 所使用刀具的螺旋角过大。 Excessive feed Excessive depth of cut Excessive long of effective length or overhang of end mill Large helix angle of flutes	降低进给速度。 降低切深量。 有效长和伸出量请调整到最低限度。 请使用螺旋角小的刀具。 Reduce feed Reduce depth of cut Adjust effective length and overhang as short as possible. Use smaller helix angle
精加工面的毛刺 Burr on Finished Surface 	切削刀刃磨损过大。 所使用刀具的螺旋角过小。 切深量太大。 Wear of cutting edge progressed. Small helix angle of flutes Excessive depth of cut	使用新的刀具或者进行返修。 请使用螺旋角大的刀具。 降低切深量。 Use new end mill or regrind Use larger helix angle Reduce depth of cut
精加工表面粗糙度不良 Poor Surface Roughness 	切削刀刃磨损过大。 卡入切屑。 进给速度太快。 有效长超出需要，伸出量太长。 主轴转速太慢。 中加工后的余量不均匀。 刀具偏摆精度不足。 Wear of cutting edge progressed Chips bite Excessive feed Excessive long of effective length or overhang of end mill Too low spindle speed Stock removals vary for finishing Excessive chucking runout	切削刀刃磨损过大。 卡入切屑。 进给速度太快。 有效长超出需要，伸出量太长。 主轴转速太慢。 中加工后的余量不均匀。 刀具偏摆精度不足。 Wear of cutting edge progressed Chips bite Excessive feed Excessive long of effective length or overhang of end mill Too low spindle speed Stock removals vary for finishing Excessive chucking runout
加工精度不良 Poor Machining Accuracy 	主轴的伸缩量不稳定。 中加工后的余量不均匀。 进给速度太快。 刀具偏摆精度不足。 Stock removals vary for finishing Excessive feed Excessive chucking runout	以使用的主轴转速进行暖机，主轴伸缩量稳定后再进行加工。 确认中加工后的余量是否均匀。 降低进给速度。 调整刀具偏摆精度。 Warm up spindle by idling before starting operation Improve semi-finishing process Reduce feed Adjust chucking runout

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